Supplementary File for: Large-Signal Stability of Power Systems with Mixtures of GFL, GFM and GSP Inverters

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Appendix A
Derivation of Model for Two-Inverter System

$$X_{\Delta 12} = \frac{X_1X_2+X_1X_g+X_2X_g}{X_g}$$
$$X_{\Delta g1} = \frac{X_1X_2+X_1X_g+X_2X_g}{X_2}$$
$$X_{\Delta g2} = \frac{X_1X_2+X_1X_g+X_2X_g}{X_1}$$
$$X_{\Sigma 1} = X_1 + X_g, X_{\Sigma 2} = X_2 + X_g$$
$$X_{1+2//g} = X_1 + \frac{X_gX_2}{X_{\Sigma 2}}, X_{2+1//g} = X_1 + \frac{X_gX_1}{X_{\Sigma 1}}$$

A. The expression of Both IBRs are GFM inverters

$$\begin{cases}\dot{\delta}_1 = k_{\text{droop1}}[P_{1ref} - \frac{V_1V_2\sin(\delta_1-\delta_2)}{X_{\Delta 12}} - \frac{V_1U_g}{X_{\Delta g1}}\sin\delta_1] \\ \dot{\delta}_2 = k_{\text{droop2}}[P_{2ref} - \frac{V_1V_2\sin(\delta_2-\delta_1)}{X_{\Delta 12}} - \frac{V_2U_g}{X_{\Delta g2}}\sin\delta_2]\end{cases} \tag{A1}$$

where δ_1, δ_2, and P_{1ref}, P_{2ref} represent the power angles, and the power references of the two GFM inverters respectively. V_1 and V_2 are the voltages of two GFM inverters.

B. The expression of Both IBRs are GFL inverters

$$\begin{cases}\dot{\delta}_1 = k_{\text{PLL1}}\left[X_{\Sigma 1}I_{d1} + X_gI_{d2}\cos(\delta_2-\delta_1) - U_g\sin\delta_1\right] \\ \dot{\delta}_2 = k_{\text{PLL1}}\left[X_{\Sigma 2}I_{d2} + X_gI_{d1}\cos(\delta_1-\delta_2) - U_g\sin\delta_2\right]\end{cases} \tag{A2}$$

where δ_1, δ_2, and I_{d1}, I_{d2} represent the angles of PLL, and the current references on d-axis of the two GFL inverters respectively.

C. The expression of IBR1 is GFL inverter and IBR2 is GFM inverter

$$\begin{cases}\dot{\delta}_1 = k_{\text{PLL}}\left[X_{1+2//g}I_{d1} - \frac{X_gV_2}{X_{\Sigma 2}}\sin(\delta_1-\delta_2) - \frac{X_2U_g}{X_{\Sigma 2}}\sin\delta_1\right] \\ \dot{\delta}_2 = k_{\text{droop}}\left[P_{ref} + \frac{X_gI_{d1}V_2}{X_{\Sigma 2}}\cos(\delta_1-\delta_2) - \frac{U_gV_2}{X_{\Sigma 2}}\sin\delta_2\right]\end{cases} \tag{A3}$$

D. The expression of IBR1 is GFL inverter and IBR2 is GSP inverter

$$\begin{cases}\dot{\delta}_1 = k_{\text{PLL1}}\left[X_{1+2//g}I_{d1} - \frac{X_gV_{ref}}{X_{\Sigma 2}}\sin(\delta_1-\delta_2) - \frac{X_2U_g}{X_{\Sigma 2}}\sin\delta_1 + \varepsilon_{lp}(k_v, k_{\text{PLL2}}, (\delta_1-\delta_2))\right] \\ \dot{\delta}_2 = k_{\text{PLL2}}\left[X_{\Sigma 2}I_{d2} + X_gI_{d1}\cos(\delta_1-\delta_2) - U_g\sin\delta_2\right]\end{cases} \tag{A4}$$

where

$$\begin{aligned}\varepsilon_{lp} &= \epsilon_v g_1(\delta_1-\delta_2) + \epsilon_{\text{PLL}}h_1(\delta_1-\delta_2) \\ g_1(\delta_1-\delta_2) &= \frac{X_g}{X_{\Sigma 2}}V_{ref}\sin(\delta_1-\delta_2) + \frac{X_g}{X_{\Sigma 2}}U_g\cos\delta_2\sin(\delta_2-\delta_1) + \frac{{X_g}^2}{X_{\Sigma 2}}I_{d1}\sin(\delta_2-\delta_1)^2 \\ h_1(\delta_1-\delta_2) &= \frac{X_g}{X_{\Sigma 2}}\cos(\delta_1-\delta_2)\end{aligned} \tag{A5}$$

Eq. (A5) gives the residual term ε_{lp}, where ϵ_{v1} is voltage control error defined as $\epsilon_{v1} \triangleq \frac{1}{k_vX_{\Sigma 2}+1}$. ϵ_{PLL} is PLL error $\epsilon_{\text{PLL}} \triangleq \dot{\delta}_2/k_{\text{PLL2}} = v_{q2}$, representing dynamics of the difference between the angle of PLL in IBR2 and its steady state. g_1 and h_1 are error functions. If k_{PLL2} is sufficiently large, implying that the PLL dynamics of the GSP inverter are faster than those of IBR1, ϵ_{PLL} can be considered negligible. Moreover, if k_v approaches infinity, ϵ_{v1} approaches zero, indicating that the GSP inverter behaves equivalently to a GFM inverter under ideal voltage regulation. Under this case ($\epsilon_{v1} \approx 0$, $\epsilon_{\text{PLL1}} \approx 0$), the sets of Equations (A3) and (A4) are identical, and they share the same form as shown in Eq. (3) in this paper.

E. The expression of IBR1 is GFM inverter and IBR2 is GSP inverter

$$\begin{cases}\dot{\delta}_1 = k_{\text{droop}}\left[P_{ref} - \frac{V_{ref}V_1\sin(\delta_1-\delta_2)}{X_{\Delta 12}} - \frac{V_1U_g\sin\delta_1}{X_{\Delta g1}} + \varepsilon_{mp}(k_v, k_{\text{PLL}}, (\delta_1-\delta_2))\right] \\ \dot{\delta}_2 = k_{\text{PLL}}\left[X_{2+1//g}I_{d2} - \frac{X_gV_1}{X_{\Sigma 1}}\sin(\delta_2-\delta_1) - \frac{X_2U_g}{X_{\Sigma 1}}\sin\delta_2\right]\end{cases} \tag{A6}$$

where

$$\begin{aligned}\varepsilon_{mp} &= \epsilon_v g_2(\delta_1-\delta_2) + \epsilon_{\text{PLL}}h_2(\delta_1-\delta_2) \\ g_2(\delta_1-\delta_2) &= -\frac{X_g}{X_{\Delta 12}X_{\Sigma 1}}{V_1}^2\cos(\delta_1-\delta_2)\sin(\delta_1-\delta_2) - \frac{X_1}{X_{\Delta 12}X_{\Sigma 1}}V_gV_1\cos\delta_2\sin(\delta_1-\delta_2) + \frac{V_1V_{ref}}{X_{\Delta 12}}\sin(\delta_1-\delta_2) \\ h_2(\delta_1-\delta_2) &= \frac{V_1\cos(\delta_1-\delta_2)}{X_{\Delta 12}}\end{aligned} \tag{A7}$$

Similarly, Eq. (A7) gives the residual term ε_{mp}, where ϵ_{v2} is voltage control error defined as $\epsilon_{v2} \triangleq 1/\big(k_vX_{2+1//g}+1\big)$. ϵ_{PLL} is PLL error, defined as above. g_2 and h_2 are the corresponding error functions. Likewise, if k_v approaches infinity, ϵ_{v2} approaches zero, indicating that the GSP inverter behaves equivalently to a GFM inverter under ideal voltage regulation.

Appendix B
Detailed Parameters in PHIL Experiment and EMT Simulation

TABLE A1
PHIL PLATFORM PARAMETERS

Parameters	Value
Base capacity S_b	60 V·A
Base voltage V_b	22.8 V
Base current I_b	2.6 A
Base impedance Z_b	8.64 Ω
Base frequency $\omega_s/2\pi$	50 Hz
Grid voltage U_g	1 pu
LC filter of IBR1 L_f	1.7 mH (0.0618 pu)
LC filter of IBR1 C_f	10 μF (0.027 pu)
Line impedance Z_1	5.5 mH + 55 mΩ (0.2j + 0.006 pu) or 13.8 mH + 100 mΩ (0.5j + 0.01 pu)
DC-link voltage of two 3-ph converters U_{dc}	54 V (2.4 pu)
Switching frequency of IBR1 (3-ph converter #1)	10 kHz
Switching frequency of power amplifier (3-ph converter #2)	50 kHz
Timestep of real-time simulation	20 μs
IBR1 control frequency	10 kHz

TABLE A2
DETAILED PARAMETERS IN GFL-GFL SYSTEM

Parameters	Value (pu)
LC Filter of IBR2 $L_f j + r_f$	$0.05\text{j} + 0.001$
LC Filter of IBR2 C_f	0.02
Inner current control loop bandwidth	1 kHz
PLL controller of IBR1 k_{PLL1}	$10 \times 2\pi$
PLL controller of IBR1 k_i	2π
PLL controller of IBR2 k_{PLL2}	$10 \times 2\pi$
PLL controller of IBR2 k_i	2π
Frequency limit of PLL ω_{limit}	± 0.2
Current reference of IBR1 I_{d1}	0.8
Current reference of IBR2 I_{d2}	0.4
Line impedance Z_1	$0.2j + 0.006$
Line impedance Z_2	$0.2j + 0.002$
Line impedance Z_g	0.35j or 0.4j
Fault resistance R_f	0.02

The EMT simulation and PHIL experiment share the same per-unit (pu) parameters.

TABLE A3
DETAILED PARAMETERS IN GFL - GFM(GSP) SYSTEM

Parameters	Value (pu)
Line mpedance Z_1	$0.5\text{j} + 0.01$
Impedance (include virtual one) Z_2	$0.1\text{j} + 0.002$
Line impedance $2Z_g$	$0.6\text{j} + 0.03$
Fault resistance R_f	0.001
Fault position (from the infinite bus)	0.8
IBR1 - GFL	
Inner current control loop bandwidth	1 kHz
PLL controller k_{PLL1}	$2.5 \times 2\pi$
PLL controller k_i	$0.2 \times 2\pi$
Frequency limit of PLL ω_{limit}	± 0.2
Current reference I_{d1}	1
IBR2 - GFM	
$p - \omega$ droop gain k_{droop}	$2.5 \times 2\pi$
Power reference P_{ref}	0.6 in Section **??**, 0 in Section **??**
AC Voltage V_2	1
Inner voltage control loop bandwidth	200 Hz
Inner current control loop bandwidth	1 kHz
LC filter of IBR2 $L_f j + r_f$	$0.2\text{j} + 0.02$
LC filter of IBR2 C_f	0.1
Current limit I_{limit}	5
$p - \omega$ droop time constant τ_p	$1/(25 \times 2\pi)$
IBR2 - GSP	
Voltage droop k_v	1 or 4 in Section **??**, 2 in Section **??**
Voltage reference V_{ref}	1
Current reference I_{d2}	0.6 in Section **??**, 0 in Section **??**
Voltage droop filter time scale τ_v	$1/(50 \times 2\pi)$
PLL controller k_{PLL2}	$k_{\text{droop}} \cdot \frac{1}{X_{\Sigma 2}}$

The EMT simulation and PHIL experiment share the same pu parameters.

TABLE A4
DETAILED PARAMETERS IN GFM - GSP SYSTEM

Parameters	Value (pu)
Line impedance Z_1	$0.5\text{j} + 0.01$
Line impedance Z_2	$0.15\text{j} + 0.002$
Line impedance $2Z_g$	$0.65\text{j} + 0.03$
Fault resistance R_f	0.001
Fault position (from the infinite bus)	0.8
IBR1 - GFM	
$p - \omega$ droop gain k_{droop}	$2.5 \times 2\pi$
Power reference P_{ref}	0.8
AC Voltage V_1	1
Inner voltage control loop bandwidth	200 Hz
Inner current control loop bandwidth	1 kHz
Current limit I_{limit}	5
$p - \omega$ droop time constant τ_p	$1/(25 \times 2\pi)$
IBR2 - GSP	
voltage control gain k_v	0, 2, or 4
Voltage reference V_{ref}	1
Current reference I_{d2}	0.2
Voltage control filter time scale τ_v	$1/(50 \times 2\pi)$
PLL controller k_{PLL2}	$6 \times 2\pi$